Exercise #1: LiPo Battery Inspection

Purpose

In this exercise, you will learn how to inspect a LiPo battery using a battery monitor. You will measure the voltage of each individual cell and the total battery voltage. Based on the measurements and a visual inspection, you will determine whether the battery is ready for use or should be charged before use.

Learning outcomes

After completing the exercise, you should be able to:

- connect a battery monitor correctly to a LiPo balance connector;
- measure the voltage of each battery cell and the total battery voltage;
- compare individual cell voltages and identify imbalance;
- estimate the battery's state of charge from its resting voltage;
- identify visible signs of damage; and
- decide whether a battery is safe and suitable for use.

Equipment

- One LiPo battery per group
- LiPo battery monitor
- Battery safety bag or designated storage container
- Calculator

General test procedure

1. Collect one LiPo battery per group and place it on a clear, dry surface.
2. Perform a visual inspection of the battery, wires, and connectors.
3. Identify the balance connector and the correct monitor orientation.
4. Carefully connect the battery monitor to the balance connector.
5. Read and record each individual cell voltage and the total battery voltage.
6. Disconnect the monitor by holding the connector housing.
7. Return the battery to the battery room once done with the tasks.

Test setup

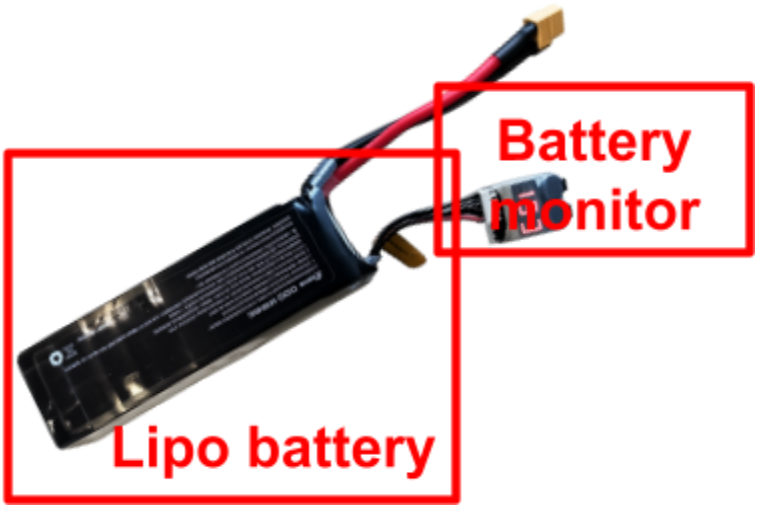


Figure 1. iPo battery with a battery monitor connected to the balance lead for checking individual cell voltages.

Safety instructions

IMPORTANT

LiPo batteries can be hazardous if they are short-circuited, physically damaged, overcharged, or excessively discharged. Handle the battery and connectors carefully throughout the exercise.

Do not connect or use the battery if you observe any of the following:

- swelling or puffing;
- damage to the battery casing;
- damaged, loose, or exposed wires;
- bent or damaged connectors;
- unusual heat;
- leakage; or
- an unusual smell.

If any of these conditions are present:

1. Do not connect the battery.
2. Place the battery in a safe area.
3. Inform the instructor immediately.

Connection safety

- Connect only to the balance connector; check the orientation and never force it.
- Keep connector pins away from metal objects.
- Disconnect by holding the connector housing; never pull on the wires.
- Do not leave the battery unattended while it is connected.

Battery information

Before taking measurements, inspect the battery label and record the following information.

Battery information	Observation
Battery type	LiPo
Number of cells	4 S
Nominal voltage	14.8 V
Capacity	5000 mAh

The number of cells is normally indicated by an S-rating. For example, 2S means two cells connected in series, while 4S means four cells connected in series. The nominal voltage of one LiPo cell is approximately 3.7 V.

Tasks

Task 1: Visual safety inspection

Complete the inspection before connecting the battery monitor.

Inspection point	Approved
The battery is not swollen or puffed	☐
The battery is not damaged	☐
The wires are not damaged or exposed	☐
The main power connector is undamaged	☐
The balance connector is undamaged	☐
The battery is not unusually warm	☐
There is no leakage or unusual smell	☐

Task 2: Measure the battery voltages

Connect the battery monitor to the balance connector and record the displayed values. Complete only the rows corresponding to the number of cells in your battery.

Measurement	Value
Cell 1 voltage	4.28 V
Cell 2 voltage	4.20 V
Cell 3 voltage	4.17 V
Cell 4 voltage	4.18 V
Cell 5 voltage	_____ V
Cell 6 voltage	_____ V
Cell 7 voltage	_____ V
Cell 8 voltage	_____ V

Verification

Add the individual cell voltages:

$V_{calculated\ total} = V_{cell\ 1} + V_{cell\ 2} + \ldots + V_{last\ cell}$

Calculated total voltage: 16.83 **V**

Task 3: Compare the cell voltages

Identify the cell with the highest and lowest measured voltages.

Highest cell voltage	4.28 V
Lowest cell voltage	4.17 V
Difference, ΔV = Vhighest − Vlowest	0.11 V

Assessment of cell balance

Difference between highest and lowest cell	General assessment
Up to approximately 0.10 V	Cells are reasonably well balanced.
Approximately 0.10–0.15 V	The battery may require balance charging.
More than approximately 0.15 V	Stop and ask the instructor to inspect the battery.

Use these values only as general guidance. Follow the instructor's decision whenever there is uncertainty.

Task 4: Estimate the battery state of charge

Battery voltage does not provide an exact capacity measurement, but the resting cell voltage can be used to make a rough estimate. First calculate the average cell voltage:

(a) Vaverage cell = Vtotal ÷ number of cells

Average cell voltage: 4.21 (4.2075) **V**

Average resting voltage per cell	Approximate condition
4.15 – 4.20 V	Fully or nearly fully charged
3.90 – 4.15 V	Partly charged
3.75 – 3.90 V	Low charged, consider charging before use
3.00 – 3.75 V	Charge before use
At or below approximately 3.0 V	Do not use; Instructor assessment required

These values are approximate and may vary with battery chemistry, temperature, age, recent loading, and manufacturer specifications.

(b) Estimated capacity (%) = ((Vtotal − Vminimum) ÷ (Vmaximum − Vminimum)) × 100

Where: Vmaximum = number of cells × 4.2 V | Vminimum = number of cells × 3.0 V

Estimated battery capacity: 100 **%**

Estimated battery condition:

- ☐ Fully or nearly fully charged
- ☐ Partly charged
- ☐ Should be charged before use
- ☐ Should be inspected by the instructor

Task 5: Make a battery decision

Combine the visual inspection, individual cell voltages, cell balance, and estimated state of charge.

Final assessment	Decision criteria
☐ **Ready for use**	The battery is undamaged, sufficiently charged, and the cells are reasonably balanced.
☐ **Charge before use**	The battery appears undamaged and reasonably balanced, but its voltage is too low for the planned flight.
☐ **Remove from service**	The battery is damaged, excessively discharged, significantly unbalanced, or otherwise unsafe.